Assignment - E

1) Show that a current loop with area A and current I in a uniform magnetic field $\vec{B}$ experiences no net magnetic force, but does experience a magnetic torque $\tau = (I\vec{A}) \times \vec{B}$

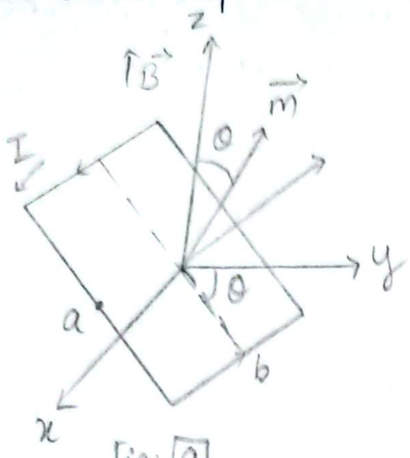


Fig: a

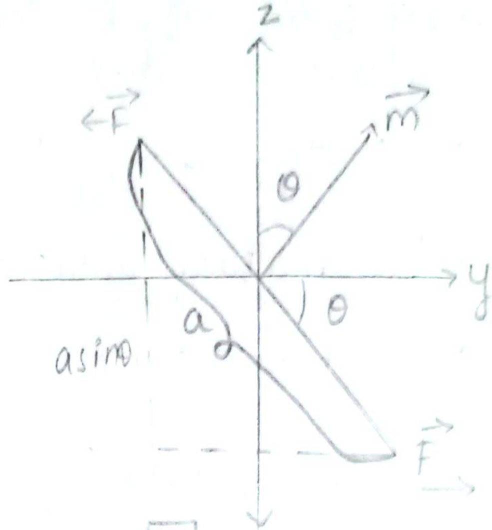


Fig: b

Consider a rectangular loop of wire carrying current I placed in a uniform magnetic field $\vec{B}$.

Let a and b be length and width of the rectangular loop and 'A' be the area defined as : $A = ab$

In a uniform field, the net force on a current carrying loop is zero:

$$\vec{F} = I \oint (d\vec{l} \times \vec{B})$$

$$= I \left(\oint d\vec{l}\right) \times \vec{B}$$

$$\therefore \vec{F} = 0$$

Opposite sides carry current in opposite direction. So,

↳ Forces on opposite direction are equal in magnitude

↳ Act in opposite directions.

In a uniform magnetic field, the net torque on a current loop is :

$$\vec{\tau} = F(a\sin\theta)\,\hat{i}$$

$$= (BIb)(a\sin\theta)\,\hat{i}$$

$$= I(ab)\,B\sin\theta\,\hat{i}$$

$$= (IA)(B\sin\theta)\,\hat{i}$$

$$\therefore \tau = (I\vec{A}) \times \vec{B} \quad [\because \vec{a} \times \vec{b} = ab\sin\theta\,\hat{n}]$$

where,

I = current flowing through loop

$\vec{A}$ = area of rectangle (loop) = ab

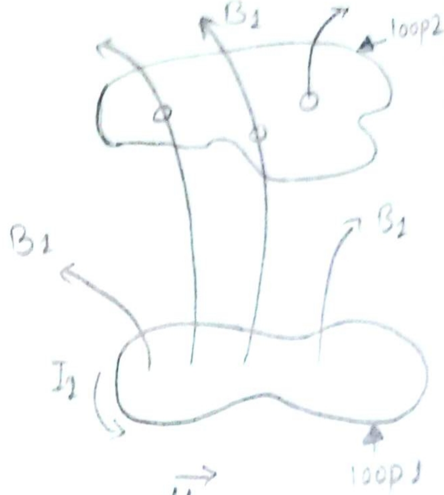
$\vec{B}$ = uniform magnetic field.

Hence, $\vec{\tau} = (I\vec{A}) \times \vec{B}$ proved

2) Define mutual inductance. Derive the Neumann formula for mutual inductance of the two loops and hence give the statement of reciprocity theorem.

↳ Mutual inductance is defined as the magnetic flux linked with one loop due to current on the other loop.

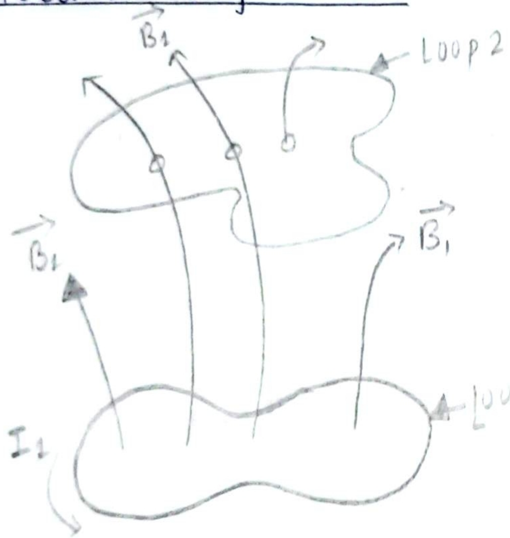
$$E_1 = M\frac{dI_1}{dt} \quad \& \quad E_2 = M\frac{dI_2}{dt}$$

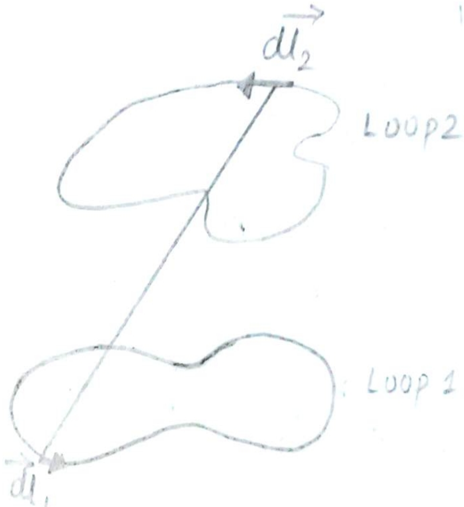


where,
M = mutual inductance of the two loops

SI unit = Henry

Neumann formula





↳ Suppose we have two loops of wire at rest. If we run a steady current I_1 around loop 1, it produces a magnetic field $\vec{B_1}$.
Some of the field lines pass through loop 2.

The Biot-Savart law,

$$\vec{B_1} = \frac{\mu_0 I_1}{4\pi}\oint\frac{\vec{dl'_1}\times\hat{\mathscr{r}}}{\mathscr{r}^2}$$

The flux of $\vec{B_1}$ through loop 2 : $\phi_2 = \int \vec{B_1}\cdot\vec{da_2}$

Thus,
$$\phi_2 = M_{21} I_1 \quad \text{--- (1)}$$

where,
M_{21} = constant of proportionality, it is known as mutual inductance of the two loops.

Again, The flux of $\vec{B}_1$ through loop 2 :

$$\phi_2 = \int \vec{B}_1 \cdot \vec{da}_2$$

$$= \int (\nabla \times \vec{A}_1) \cdot \vec{da}_2 \quad \text{[using Stoke's theorem]}$$

$$= \oint_{C2} \vec{A}_1 \cdot \vec{dl}_2$$

$$= \oint_{C2} \left[\frac{\mu_0 I_1}{4\pi} \oint_{C1} \frac{\vec{dl}_1}{r} \right] \cdot \vec{dl}_2$$

$$\therefore \phi_2 = \left[\frac{\mu_0}{4\pi} \oint_{C2} \oint_{C1} \frac{\vec{dl}_1 \cdot \vec{dl}_2}{r} \right] I_1 \quad — (2)$$

Comparing (1) & (2), we get,

$$M_{21} = \frac{\mu_0}{4\pi} \oint_{C2} \oint_{C1} \frac{\vec{dl}_1 \cdot \vec{dl}_2}{r} \quad \text{[Neumann Formula]}$$

Reciprocity Theorem

Neumann Formula :

$$M_{21} = \frac{\mu_0}{4\pi} \oint_{C2} \oint_{C1} \frac{\vec{dl}_1 \cdot \vec{dl}_2}{r} \quad — (3)$$

It's not very useful for practical purposes, but it does reveal two important things about mutual inductance:

(1) M_{21} is purely geometrical quantity, having to do with the sizes, shapes and relative positions of the two loops.

2) The integral in Eq^n (3) is unchanged if we switch the roles of loops 1 and 2; it follows that

$$M_{21} = M_{12} \qquad \because \left[\phi_1 = \left[\frac{\mu_0}{4\pi} \oint_{C1} \oint_{C2} \frac{\vec{dl}_2 \cdot \vec{dl}_1}{r} \right] I_2 \right]$$

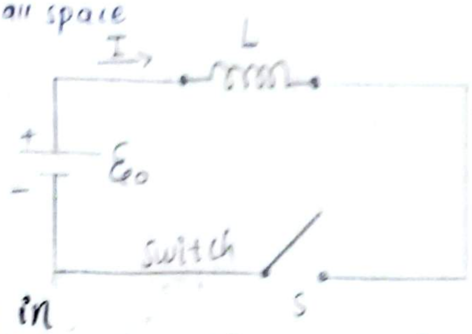
This is the reciprocity theorem.

Whatever the shapes and positions of the loops, the flux through 2 when we run a current around 1 is identical to the flux through 1 when we send same current around 2.

$$\phi_2 = \left[\frac{\mu_0}{4\pi} \oint_{C2} \oint_{C1} \frac{\vec{dl}_1 \cdot \vec{dl}_2}{r} \right] I = \left[\frac{\mu_0}{4\pi} \oint_{C1} \oint_{C2} \frac{\vec{dl}_2 \cdot \vec{dl}_1}{r} \right] I = \phi_1$$

3) Show that the energy stored in a magnetic field $\vec{B}$ is given by $W = \frac{1}{2\mu_0}\int_{all\ space} B^2\, d\tau$.

4 Figure shows a circuit in which a coil is connected in series with a battery.



As soon as the switch S is closed, the current starts rising in the circuit. The back emf produced in the coil opposes this rise in current. So, work must be done against the back emf to get the current going.

The work done on a unit charge, against the back emf, in one trip around the circuit is $-\mathcal{E}$.

The amount of charge per unit time passing down the wire is I

So, the total work done per unit time is

$$\frac{dW}{dt} = -\mathcal{E} I = LI\frac{dI}{dt}$$

If we start with zero current and build it up to a final value I, the total work done is

$$W = \int_0^I LI\frac{dI}{dt}\, dt = L\int_0^I I\, dI$$

$$\therefore W = \frac{1}{2}LI^2 \quad \text{--- (1)}$$

$$= \frac{1}{2}I(LI)$$

$$= \frac{1}{2}I\phi \quad [\because \text{the flux through the loop, } \phi = LI]$$

$$= \frac{1}{2}I\int_S \vec{B}\cdot d\vec{a}$$

$$= \frac{1}{2}I\int_S (\nabla\times\vec{A})\cdot d\vec{a}$$

$$= \frac{1}{2}I\oint_c \vec{A}\cdot d\vec{l}$$

$$\therefore W = \frac{1}{2}\oint_c (\vec{A}\cdot\vec{I})\, dl \quad \text{--- (2)}$$

In this form, the generalization to volume currents is obvious:

$$W = \frac{1}{2}\int_V (\vec{A}\cdot\vec{J})\,d\tau$$

$$= \frac{1}{2\mu_0}\int_V \vec{A}\cdot(\nabla\times\vec{B})\,d\tau \qquad [\because \text{Ampere's law, } \nabla\times\vec{B} = \mu_0\vec{J}]$$

$$= \frac{1}{2\mu_0}\int_V [\vec{B}\cdot(\nabla\times\vec{A}) - \nabla\cdot(\vec{A}\times\vec{B})]\,d\tau \quad [\because \nabla\cdot(\vec{A}\times\vec{B}) = \vec{B}\cdot(\nabla\times\vec{A}) - \vec{A}\cdot(\nabla\times\vec{B})]$$

$$= \frac{1}{2\mu_0}\left[\int_V B^2\,d\tau - \int_V \nabla\cdot(\vec{A}\times\vec{B})\,d\tau\right]$$

$$= \frac{1}{2\mu_0}\left[\int_V B^2\,d\tau - \oint_S (\vec{A}\times\vec{B})\cdot d\vec{a}\right] \quad \text{--- (3)}$$

where S is the surface bonding the volume V.

In particular, if we integrate over all space, then the surface integral goes to zero and we are left with

$$\therefore W = \frac{1}{2\mu_0}\int_{\text{all space}} B^2\,d\tau$$

Thus, the energy stored per unit volume (i.e. energy density) is equal to

$$U_B = \frac{B^2}{2\mu_0}$$